Shubham Kumar Singh

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Expected graduation: January 2027 | **Availability:** full-time 13 to 17 week internship in London, Summer 2027
Right to work: UK Student visa; will apply for the Graduate visa on graduation (no sponsorship needed for the internship)

Profile

MSc AI & Computer Science student at Northeastern University London with 3+ years in IT support and operations at Capgemini and Tech Mahindra: incident triage against SLA, troubleshooting and Python automation. Builds and measures LLM systems: a live RAG assistant and an LLM tool used daily in production. Seeking a Software Engineering or Site Reliability Engineering internship.

Education

MSc Artificial Intelligence & Computer Science | Northeastern University London — Jan 2026 - Jan 2027
- Dissertation: security guardrails for LLM agents (does an agent split a blocked refund into smaller ones that pass?)
- Modules include Web Services (84%), Programming for Data Applications (72%) and Programming Design Paradigm (70%).

Advanced Certificate Programme in Data Science (NLP) | IIIT Bangalore with upGrad | CGPA 3.85/4.0 — Jan 2025 - Oct 2025

Bachelor of Computer Applications | Chandigarh University, India — Sep 2021 - Sep 2024

Diploma in Computer Science & Engineering | Kalinga University, India | 81% — 2018 - 2021

Experience

Campus Ambassador (part-time) | Northeastern University London, London, UK — Aug 2026 - Present
- Keep open days running by setting up projectors, microphones and laptops and troubleshooting AV and IT issues on the day.

Senior IT Support Analyst | Capgemini, Noida, India — Jun 2023 - Dec 2025
- Cut 2 to 5 hours of manual toil a week, with a human approving every change before it reached the directory, by building a Python tool that used an internal LLM to convert prose access-request tickets into proposed Active Directory changes.
- Met SLA targets consistently on 50+ incidents a day for two and a half years, limiting downtime on business-critical systems, by triaging the ServiceNow queue and troubleshooting faults remotely with Bomgar.
- Wrote Python scripts to automate routine incident reporting, administered accounts, access and endpoint compliance in Active Directory and Tanium, and onboarded and mentored new team members.

Technical Support Associate | Tech Mahindra (IBU TAC), Noida, India — Aug 2022 - Jun 2023
- Resolved complex Netgear and Orbi network configuration and connectivity faults, escalations included, with consistently positive customer feedback, by troubleshooting independently and following up every case in Salesforce CRM within SLA.

Customer Support Associate | Tech Mahindra, India — Aug 2021 - May 2022
- Earned consistently positive customer feedback and supervisor recognition for reliability in a high-volume contact centre by resolving JioMart e-commerce cases and keeping accurate records in Kapture CRM.

Projects

Portfolio Assistant: live RAG app | *TypeScript*, *Next.js*, *Gemini*, *Voyage*, *Groq* — github.com/Shubham-afk-del/portfolio-assistant
- Kept the live app within free-tier limits (about 200 questions a day), with embedding waits capped at 2.5 s, by adding rate limits, graceful degradation to keyword search, failover across three models, unit tests and CI on GitHub Actions.
- Shipped portfolio-assistant-iota.vercel.app, which answers recruiters' questions with the sources shown, passing 77 of 86 golden questions in 10 categories including prompt injection, by building a RAG pipeline over 238 passages of my work.
- Raised recall@10 from 0.951 with dense search alone to 0.985 on 34 labelled questions by adding a hand-written BM25 ranker, reciprocal rank fusion and an LLM reranker that grades the top 20 passages.

Fraudulent Claim Detection | *Python*, *scikit-learn*, *statsmodels* — github.com/Shubham-afk-del/Fraudulent-Claim-Detection
- Caught 84% of held-out frauds at 85% specificity (F1 0.729, AUC 0.855) on 1,000 claims by building a six-feature logistic regression (RFECV, VIF checks, cutoff selection) that beat an overfit random forest; flagged a label leak in the synthetic data.

Delivery-Time Prediction | *Python*, *pandas*, *scikit-learn* — github.com/Shubham-afk-del/Delivery-Time-Prediction
- Predicted delivery times for 175,777 orders (test R-squared 0.880, MAE 2.33 min) by engineering 12 features for a linear regression; showed that RFE down to 8 features made it worse and traced the root cause to unstable feature selection.

Also on GitHub: employee-retention model (AUC 0.825), NYC taxi EDA (489,542 trips), 29 MySQL queries with window functions.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL (MySQL, SQLite), C++ (familiar); data structures and algorithms practice in Python, logged in a public TIL repo
AI & machine learning: RAG and information retrieval (BM25, dense retrieval, reciprocal rank fusion, LLM reranking), LLM evaluation, agentic AI, scikit-learn, statsmodels, pandas, NumPy; familiar with PyTorch, TensorFlow and NLP
Engineering: Git, GitHub Actions (CI), Next.js, SvelteKit, REST APIs, Vercel; familiar with Docker and AWS
Enterprise IT: ServiceNow, Active Directory, Tanium, Bomgar, Salesforce CRM; incident management, network troubleshooting

Certifications

Foundations of Responsible AI (Northeastern, 2026); AWS Cloud Technical Essentials (2025); Google Foundations of Cybersecurity (2025); upGrad Python Bootcamp (2025). **In progress:** AI Engineer Agentic Track: The Complete Agent & MCP Course (Udemy)